TABLE 2
STAGE 1-POWER SWITCHING OPTIONS

	Options					
Input	48V, 64A Power Switching	ASR-SI240D60ZW-M (₹2,783)	A31ACQ24VDC1 (₹150)	3-1414937-3 (₹741)	CB1AH-P-12V (₹298)	A3F1ACQ12VDC1D (₹155)
						A31UCQ12VDC1R (₹160)
	Output (Switching Current)	7.5mA	75mA	83mA	117mA	150mA
	Return (₹)	2,783	150	741	298	155

tion does away with that assumption and instead finds out the interest rate that makes NPV equal to zero. The RoI calculation is quite complex and is best done using an inbuilt function in spreadsheets or a special calculator that has such functions. Using the values from Table 1, we can find the RoI. This comes out as 582%. This value looks too good to be true. We shall revise our analysis once we gather more details about our product.

Profit margin. While making a product the money we spend is called cost. The cost of the product is a result of our decisions. We cannot change it. While we sell the product, we decide to sell it at a price. Price is our decision, and we can change it quickly. The difference between price and cost is our profit.

Profit margin is the ratio of profit over price. We use it to compare how our product compares with others in the market. Innovative products tend to have a higher profit margin, whereas products that have competition have a lower profit margin.

In our case, our target price is ₹1,000 whereas our product cost is estimated to be ₹600. So our profit margin is (1,000-600)/1,000 = 0.40. For our product, which we want to make very innovative, 40% profit margin may be justified for the initial period.

Breakeven quantity. There are many unpredictable things in product development. One must invest a fixed amount in developing the product and purchasing machinery and equipment. Once we start selling the product, we start recovering our investment.

In the development expenses, there are two parts. One part is the fixed cost. These are the one-time investments in machines, project costs, etc. Another part is variable, which varies with the number of products we make. Costs of raw materials and wages of workers fall in this category.

Breakeven quantity is the number/quantity of the product we must sell to recover our investment. Numerically, it is Fixed Cost/Profit per item. For our charge controller, we need to invest ₹5.5 million. Our profit per item is ₹400. So, we need to sell 13,750 pcs of the charge controller to recover our investment.

The breakeven quantity gives us an indication of the risk of our investment. If the breakeven quantity is very big then we run a bigger risk.

Payback period. While breakeven quantity gives the number of items to be sold to make a profit, we also need to know how much time it will take before we see profit from our work. The payback period gives us that number. Payback period is the ratio of breakeven quantity and per day production.

We plan to make 100 pieces per day. Hence our payback period is 13,750/100 = 137.5 days.

These economic parameters give us an indication of the profitability and risk of our product development work. Ideas that give more profit with less risk are always preferable.

Risk assessment

Apart from economic analysis, we should also do a risk assessment of our new product development. Risks

TABLE 3
STAGE 2-CONTROL CURRENT

Switching Current		Best Return (₹)	
Inputs	7.5mA	2,783	Direct ₹2,783
	75mA	150	Transistor BC148, flyback diode ₹170
	83mA	741	Transistor BC148, flyback diode ₹761
	117mA	298	IRFZ44N, flyback diode ₹328
	150mA	155	IRFZ44N, flyback diode ₹185
	Control current		15mA

for a new product come from changes in the sales volume, different prices, and periods. Changes in government policies can also cause changes in our original plan. Another risk is competition.

As we develop new products, our competition may also come up with similar or better products. During product development, we need to maintain confidentiality to protect our competitive lead. Our sales projection should consider the effects of future competition.

During product viability analysis one should see the effects of both optimistic as well as conservative estimates and decide if the product development efforts look viable.

Selecting alternatives

Product design involves selecting the right design from different alternatives.